CLOUD SECURITY — PRACTICAL-BASED LEARNING

PBL 5: Configure AWS CloudTrail and Store Audit Logs in S3

Standalone report • Anticipated troubleshooting scenarios clearly labeled

1. Aim

Configure AWS CloudTrail to record AWS API management activity and deliver audit logs to an Amazon S3 bucket. Verify evidence of EC2 instance creation, EC2 instance termination and IAM user creation.

2. Problem Statement

Create a dedicated S3 bucket for CloudTrail logs, configure a trail to record management events including write events, configure log delivery and permissions, generate the required test events, and verify them in CloudTrail Event history and in S3.

3. Tools and Services

- AWS Management Console
- AWS CloudTrail
- Amazon S3
- Amazon EC2
- AWS IAM
- CloudTrail Event history and S3 object listing

4. Procedure

Step 1 — Create a dedicated S3 bucket

Create a globally unique S3 bucket dedicated to audit logs. Choose the appropriate AWS Region and keep public access blocked. Do not use a bucket containing unrelated personal or application data.

Step 2 — Create a CloudTrail trail

Open CloudTrail and create a trail. Configure it to record management events, including write events, and select the dedicated S3 bucket as the log destination. Enable multi-Region logging if required by the lab or chosen design.

Step 3 — Configure log delivery and permissions

Use the CloudTrail setup workflow to configure the bucket policy and delivery permissions. Verify that the trail status is logging and that the destination bucket is selected correctly.

Step 4 — Generate test events

Using an authorized lab account, launch a test EC2 instance, then terminate it after capturing the required evidence. Create a test IAM user only if permitted by the lab account and its authorization rules. Use least privilege and avoid creating unnecessary identities.

Step 5 — Find events in CloudTrail Event history

Search the appropriate Region and time window for the EC2 creation event, EC2 termination event and IAM user creation event. Record event name, event time, identity and affected resource.

Step 6 — Verify S3 log delivery

Open the configured S3 bucket and locate the CloudTrail log objects under the expected AWS log prefix. Delivery can take time; do not assume immediate appearance. If needed, inspect the log contents for the expected event records.

5. Anticipated Errors and Solutions

The following are common possible issues to prepare for. They are anticipated scenarios, not claims that these errors occurred in this user's lab.

Error 1 — S3 bucket name is unavailable

S3 bucket names must be globally unique. Choose a new unique name and retry.

Error 2 — CloudTrail cannot deliver logs to S3

Check that the selected bucket is correct, the bucket policy permits CloudTrail delivery, and the trail and bucket are configured in the intended account and Region. Prefer the CloudTrail console workflow to create the required policy rather than guessing permissions.

Error 3 — Trail is not logging

Check the trail status, selected event types, Region settings and destination. Save the configuration and confirm logging is enabled.

Error 4 — Events do not appear in Event history

Confirm the correct Region and time range, search by event name, and verify that the action actually occurred. Event history is Region-scoped for many events. Check that the account has permissions to view event history.

Error 5 — Logs are not visible in S3 immediately

CloudTrail delivery is not instantaneous. Wait, refresh the bucket listing, verify the trail's destination and delivery status, and inspect the expected AWS log prefix. A missing object immediately after an event does not by itself prove configuration failure.

Error 6 — Access denied while creating or terminating EC2 resources

The lab identity may lack required permissions or an organization policy may deny the action. Use an authorized lab role with only the permissions required for the exercise; do not bypass organizational controls.

Error 7 — IAM user creation is denied

Check whether the lab account permits IAM user creation and whether the current identity has the necessary authorized permission. If IAM user creation is prohibited, do not attempt to bypass the restriction; document the limitation and ask the instructor or account administrator for an approved test environment.

Error 8 — S3 log objects are inaccessible

Check IAM permissions, bucket policy, encryption settings and the selected account. If a customer-managed KMS key is used, ensure the authorized CloudTrail and reader permissions are configured correctly. Do not make the bucket public as a workaround.

Error 9 — Unexpected AWS charges or leftover resources

Check EC2 instances, volumes, Elastic IPs, trails and S3 storage after testing. Terminate the test instance and remove only disposable lab resources that are no longer needed. Preserve required audit evidence before cleanup.

6. Result and Verification Status

Expected result: CloudTrail records the required management events and delivers audit log files to the dedicated S3 bucket. Event history and the S3 objects provide separate evidence: Event history confirms searchable events, while S3 object inspection confirms log delivery.

No actual execution evidence was provided for PBL 5 in the available conversation. Therefore, this report describes the required procedure and anticipated troubleshooting only; it does not claim that a trail, test events or S3 delivery have already been verified.

7. Security Measures

- Block all public access to the audit-log bucket.
- Use a dedicated bucket and least-privilege permissions.
- Enable S3 encryption and consider versioning or retention controls according to lab requirements.
- Record management events including write events, as required by the assignment.
- Do not store credentials or unnecessary personal information in screenshots or test logs.
- Preserve evidence before terminating disposable resources.
- Monitor AWS costs and clean up test resources safely.

8. Conclusion

This practical demonstrates how CloudTrail can record management activity and deliver audit logs to S3. Verification requires evidence of the EC2 creation event, EC2 termination event and IAM user creation event, along with the corresponding delivered log objects. The errors listed in this report are anticipated troubleshooting scenarios and must not be presented as errors actually encountered unless they occur during the lab.

9. Screenshots Checklist

- ■ Dedicated S3 bucket details with Block Public Access enabled.
- ■ CloudTrail trail configuration and selected S3 destination.
- ■ Management event settings, including write events.
- ■ Trail status showing logging.
- ■ CloudTrail Event history for EC2 instance creation.
- ■ CloudTrail Event history for EC2 instance termination.
- ■ CloudTrail Event history for IAM user creation.
- ■ S3 bucket showing delivered CloudTrail log objects.
- ■ Relevant event record details: event name, timestamp, identity and resource.
- ■ Test EC2 instance terminated after evidence collection.